Failure Tolerance in High-Dimensional TuRBO

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Abstract

Trust-region Bayesian optimization methods maintain a center x_k, a local region $\mathcal{T}_k$, and a length parameter Δ_k. The length parameter is shrunk when the local failure counter reaches a prescribed tolerance. In the reference implementation of TuRBO [2, 3], the number $K_{k,c}$ of active RAASP mask coordinates satisfies

$$\mathbb{E}_D[K_{k,c}] = 20 + e^{-20} + O(D^{-1}), \qquad D \to \infty.$$

The failure tolerance is ambient-dimensional: a shrink can be triggered only after at least D evaluations are assigned through unsuccessful updates in the same region. We formulate this scale separation in trust-region notation and prove

$$N_\downarrow(T) \le \left\lfloor \frac{T}{D} \right\rfloor,$$

where $N_\downarrow(T)$ is the number of possible shrink events after T assigned evaluations. Hence $T < D$ implies $N_\downarrow(T) = 0$, even along an all-failure trajectory.

1 Trust-Region State

Let

$$\mathcal{X}_D = [0,1]^D, \qquad \mathcal{D}^D_{++} = \{\, \mathrm{diag}(w_1, \ldots, w_D) : w_j > 0 \,\}.$$

At local iteration k, write the trust-region state as

$$\mathsf{S}_k = (x_k, \Delta_k, W_k) \in \mathcal{X}_D \times (0,\infty) \times \mathcal{D}^D_{++}. \tag{1.1}$$

The scalar Δ_k is the update variable; W_k fixes relative coordinate scales. The induced weighted ℓ_∞ region is

$$\mathcal{T}_k = \mathcal{T}(\mathsf{S}_k) = \left\{ x \in \mathcal{X}_D : \left\| W_k^{-1}(x - x_k) \right\|_\infty \le \frac{\Delta_k}{2} \right\}. \tag{1.2}$$

This is the standard trust-region object, specialized to the axis-aligned hyperrectangles used by TuRBO [1, 2].

In TuRBO, W_k is determined by local GP lengthscales. For $\ell_k^{\mathrm{GP}} \in (0,\infty)^D$, define

$$g_k = \left(\prod_{h=1}^{D} \ell^{\mathrm{GP}}_{k,h} \right)^{\frac{1}{D}}, \qquad w_{k,j} = \frac{\ell^{\mathrm{GP}}_{k,j}}{g_k}, \qquad W_k = \mathrm{diag}(w_{k,1}, \ldots, w_{k,D}). \tag{1.3}$$

Then

$$\det W_k = 1, \qquad L_{k,j} = \Delta_k w_{k,j}, \qquad \mathrm{GM}(L_{k,1}, \ldots, L_{k,D}) = \Delta_k. \tag{1.4}$$

Away from boundary clipping, $\mathrm{Vol}(\mathcal{T}_k) = \prod_{j=1}^{D} L_{k,j} = \Delta_k^D$. ENN-surrogate TuRBO uses the isotropic specialization $W_k = I_D$. Hence $L_{k,j} = \Delta_k$ for every coordinate.

2 RAASP Candidate Support

TuRBO generates a candidate cloud inside $\mathcal{T}_k$ by applying a random axis-aligned subspace perturbation (RAASP) to a base trust-region displacement. Write

$$[D] = \{1, \ldots, D\}, \qquad e_j = (0, \ldots, 0, 1, 0, \ldots, 0) \in \{0,1\}^D.$$

For candidate c, let

$$u_{k,c} \in \left[-\frac{\Delta_k}{2}, \frac{\Delta_k}{2}\right]^D, \tag{2.1}$$

$$\bar{b}_{k,c,j} \overset{\text{ind}}{\sim} \text{Bernoulli}(p_D), \qquad j \in [D], \tag{2.2}$$

$$p_D = \min\left\{\frac{20}{D}, 1\right\}. \tag{2.3}$$

Set

$$\bar{Z}_{k,c} = \sum_{j=1}^{D} \bar{b}_{k,c,j}. \tag{2.4}$$

The reference implementation then enforces a nonzero mask. Equivalently, for an independent $J_{k,c} \sim \text{Unif}([D])$,

$$b_{k,c} = \begin{cases} \bar{b}_{k,c}, & \bar{Z}_{k,c} > 0, \\ e_{J_{k,c}}, & \bar{Z}_{k,c} = 0. \end{cases} \tag{2.5}$$

This is the RAASP correction used in the reference code [4]. Let $\mathbb{P}_D$ denote the joint law of $(\bar{b}_{k,c}, J_{k,c})$, and let $\mathbb{E}_D$ denote expectation under $\mathbb{P}_D$. The proposed and realized steps below are conditional on the trust-region state S_k. The proposed and realized steps are

$$\tilde{s}_{k,c} = W_k(b_{k,c} \odot u_{k,c}), \tag{2.6}$$

$$x_{k,c} = \Pi_{\mathcal{X}_D}(x_k + \tilde{s}_{k,c}), \qquad s_{k,c} = x_{k,c} - x_k. \tag{2.7}$$

Thus $\tilde{s}_{k,c}$ is the proposed step before clipping, and $s_{k,c}$ is the realized step after projection onto $\mathcal{X}_D$. Boundary clipping can only remove active coordinates:

$$\text{supp}(s_{k,c}) \subseteq \text{supp}(\tilde{s}_{k,c}), \qquad \|s_{k,c}\|_0 \le \|\tilde{s}_{k,c}\|_0. \tag{2.8}$$

For $s \in \mathbb{R}^D$, define

$$A(s) = \text{supp}(s) = \{j : s_j \neq 0\}, \qquad \|s\|_0 = |A(s)|. \tag{2.9}$$

The mask support and its scale are

$$K_{k,c} = \|b_{k,c}\|_0, \qquad \kappa_D = \mathbb{E}_D[K_{k,c}]. \tag{2.10}$$

If $u_{k,c,j} \neq 0$ almost surely for every j, then

$$A(\tilde{s}_{k,c}) = \{j : b_{k,c,j} = 1\}, \qquad \|\tilde{s}_{k,c}\|_0 = K_{k,c}, \tag{2.11}$$

because W_k has strictly positive diagonal entries. The generator is coordinate-sparse when $\kappa_D = o(D)$, and constant-coordinate when $\kappa_D = O(1)$.

Lemma 1 (Sparse-coordinate scale)**.** *Let K_D denote a generic draw from the distribution of $K_{k,c}$. Then*

$$K_D \overset{d}{=} \max\{1, Z_D\}, \qquad Z_D \sim \text{Binomial}(D, p_D), \tag{2.12}$$

and

$$\mathbb{E}_D[K_D] = Dp_D + (1 - p_D)^D. \tag{2.13}$$

For $D > 20$,

$$\mathbb{E}_D[K_D] = 20 + \left(1 - \frac{20}{D}\right)^D = 20 + e^{-20} + O(D^{-1}). \tag{2.14}$$

Equivalently,

$$\kappa_D = 20 + e^{-20} + O(D^{-1}), \qquad \frac{\kappa_D}{D} \to 0.$$

Proof. Before correction,

$$Z_D = \sum_{j=1}^{D} \bar{b}_{k,c,j} \sim \text{Binomial}(D, p_D).$$

The correction changes the cardinality only on $\{Z_D = 0\}$, where it sets the cardinality to one. Therefore $K_D = \max\{1, Z_D\}$, and

$$\mathbb{E}_D[K_D] = \mathbb{E}_D[Z_D] + \mathbb{P}_D\{Z_D = 0\} = Dp_D + (1 - p_D)^D.$$

Substituting $p_D = \frac{20}{D}$ gives (2.14). $\square$

ENN-surrogate `TuRBO` uses the equivalent support-size representation: draw $K_D = \max\{1, Z_D\}$, then choose K_D coordinates uniformly without replacement. The support-cardinality law is the same as in (2.5).

Thus the candidate generator has active-coordinate scale $\kappa_D = 20 + e^{-20} + O(D^{-1}) = o(D)$. The next section shows that the failure tolerance has scale at least D.

3 Failure Counter and Shrink Rule

Now separate the trust-region update rule from the particular surrogate. The surrogate and acquisition function decide which evaluations are successes; the following counting argument only uses the resulting success/failure sequence. Fix one region. Let

$$C_k \in \mathbb{N}_0, \qquad a_k \in \mathbb{N}, \qquad \eta_k \in \{0, \ldots, a_k\}, \qquad I_k \in \{0, 1\},$$

where C_k is the failure counter before update k, a_k is the number of evaluations assigned to the region at update k, η_k is the evaluation-normalized failure increment used by the implementation, and $I_k = 1$ denotes a successful update. Define the pre-shrink counter

$$C_{k+1}^{-} = (1 - I_k)(C_k + \eta_k). \tag{3.1}$$

Given a failure tolerance τ_D, the shrink indicator is

$$H_k = \mathbf{1}\left\{C_{k+1}^{-} \geq \tau_D\right\}. \tag{3.2}$$

For a shrink factor $\gamma_\downarrow \in (0, 1)$, the length update is

$$\Delta_{k+1} = \gamma_\downarrow^{H_k} \Delta_k. \tag{3.3}$$

Thus $H_k = 1$ shrinks the region and $H_k = 0$ leaves Δ_k unchanged. The counter reset is

$$C_{k+1} = (1 - H_k)C_{k+1}^{-}. \tag{3.4}$$

Thus τ_D is the failure tolerance, measured in evaluations assigned to the region. This is the scale to compare against κ_D.

Theorem 1 (Shrink-count bound)**.** *Consider any counter-based trust-region method satisfying $\tau_D \geq D$. Let T be the number of evaluations assigned to a fixed region after initialization. Put*

$$A_0 = 0, \qquad A_K = \sum_{k=1}^{K} a_k, \qquad K(T) = \max\{K \in \mathbb{N}_0 : A_K \leq T\}, \qquad N_\downarrow(T) = \sum_{k=1}^{K(T)} H_k. \tag{3.5}$$

Then

$$N_\downarrow(T) \leq \left\lfloor \frac{T}{\tau_D} \right\rfloor \leq \left\lfloor \frac{T}{D} \right\rfloor. \tag{3.6}$$

Proof. Let

$$F_T = \sum_{k=1}^{K(T)} (1 - I_k)\eta_k.$$

Then

$$F_T \le \sum_{k=1}^{K(T)} a_k = A_{K(T)} \le T.$$

Every shrink event requires failure increments with total mass at least τ_D. By (3.4), the counter is reset after each shrink, so the increments certifying distinct shrink events are disjoint. Hence

$$N_\downarrow(T)\tau_D \le F_T \le T.$$

The first inequality in (3.6) follows. The second follows from $\tau_D \ge D$. □

Corollary 1 (No shrink below ambient budget)**.** *If $T < D$, then $N_\downarrow(T) = 0$. Thus no failure-driven shrink can occur, even if every update is unsuccessful.*

4 `TuRBO` Failure Tolerance

We now identify τ_D for the reference `TuRBO` rules. This is the only place where implementation details enter the argument. The official code uses the minimization convention: a batch is successful if it improves the local incumbent by the implementation tolerance

$$y_k^{\min} = \min_{1\le i\le q} f(x_{k,i}), \qquad I_k = 1 \quad\Longleftrightarrow\quad y_k^{\min} < f_{\text{best}} - 10^{-3}|f_{\text{best}}|. \tag{4.1}$$

The exact success criterion only determines when the counter resets; the bound above holds for every success/failure sequence.

`Turbo1`. With batch size q, the reference single-region code uses the batch-level failure tolerance [4]

$$\widehat{\tau}_D^{(1)}(q) = \left\lceil \max\left\{\frac{4}{q}, \frac{D}{q}\right\}\right\rceil. \tag{4.2}$$

In evaluation units this is

$$\tau_D^{(1)}(q) = q\widehat{\tau}_D^{(1)}(q) \ge D \tag{4.3}$$

Thus one shrink requires at least $\tau_D^{(1)}(q)$ evaluations assigned through unsuccessful batches.

ENN-surrogate `TuRBO`. In ENN-surrogate `TuRBO` with batch size q, the trust-region failure clock is batch-level:

$$\widehat{\tau}_D^{(\mathrm{ENN})}(q) = \left\lceil \max\left\{\frac{4}{q}, \frac{D}{q}\right\}\right\rceil. \tag{4.4}$$

Under full-batch updates, each unsuccessful update assigns q evaluations to the region. The evaluation threshold is

$$\tau_D^{(\mathrm{ENN})}(q) = q\widehat{\tau}_D^{(\mathrm{ENN})}(q) \ge D. \tag{4.5}$$

The shrink-count bound therefore applies unchanged to ENN-surrogate `TuRBO` in the full-batch regime.

`TurboM`. The reference multi-region code counts failures by evaluations assigned to each region and uses [5]

$$\tau_D^{(m)} = \max\{5, D\}. \tag{4.6}$$

Therefore $\tau_D^{(m)} = D$ for $D \ge 5$.

Corollary 2 (`TuRBO` shrink bound)**.** *For `Turbo1`, ENN-surrogate `TuRBO` under full-batch updates, and `TurboM`, in dimensions $D \ge 5$, a trust region receiving T post-initialization evaluations satisfies*

$$N_\downarrow(T) \le \left\lfloor \frac{T}{D} \right\rfloor. \tag{4.7}$$

5 Consequences

The reference rule shrinks by $\gamma_\downarrow = \frac{1}{2}$. If a region starts at Δ_0 and restarts once $\Delta_k < \Delta_{\min}$, then the number of shrink events required for one restart is

$$h_{\text{restart}} = \min\{h \in \mathbb{N} : \Delta_0 2^{-h} < \Delta_{\min}\} = \left\lfloor \log_2\left(\frac{\Delta_0}{\Delta_{\min}}\right)\right\rfloor + 1. \tag{5.1}$$

With the reference defaults $\Delta_0 = 0.8$ and $\Delta_{\min} = 2^{-7}$,

$$h_{\text{restart}} = 7, \qquad T_{\text{restart}} \geq h_{\text{restart}}\tau_D \geq 7D.$$

Thus a default restart requires at least $7D$ evaluations assigned through unsuccessful updates in the same region.

The mathematical point is the scale separation

$$\kappa_D = \mathbb{E}_D[K_{k,c}] = 20 + e^{-20} + O(D^{-1}), \qquad \tau_D \geq D. \tag{5.2}$$

Equivalently,

$$\frac{\kappa_D}{D} \to 0, \qquad \frac{\tau_D}{D} \geq 1, \qquad \frac{\tau_D}{\kappa_D} \to \infty. \tag{5.3}$$

The proposed candidate step is low-support in ambient coordinates; the realized step after clipping has no larger support by (2.8). The length update, however, waits for an ambient-dimensional number of assigned evaluations in unsuccessful updates. Hence the failure-driven length update cannot be triggered whenever the per-region budget T is below D. This is not a claim that `TuRBO` is ineffective in all high-dimensional problems; it is a statement about the ambient-dimensional failure tolerance in the regime $T < D$.

Example 1. If $D = 100{,}000$ and a region receives $T = 10{,}000$ evaluations, then $N_\downarrow(T) = 0$ under the reference rule, even along an all-failure trajectory. A default restart requires on the order of 700,000 assigned evaluations in unsuccessful updates in that same region.

Remark 1. The ENN surrogate changes the local model and candidate ranking, but not the two scales isolated above: the RAASP support remains constant-coordinate, and under full-batch updates the failure-driven length update remains ambient-dimensional.

References

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